Empirical Function Method: A Precise Approach for Filling Data Gaps in Satellite Sea Surface Temperature Imagery

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***Abstract*— The acquisition of wide geophysical data of vast oceans by satellites can be impeded by clouds, which may result in gaps in the data acquired in the infrared and visual bands, such as sea surface temperature (SST) data, leading to limited data usage. To address this issue, we proposed a general and robust method, named the empirical function (EF) method, which involves expanding an ocean field with space-time-separation functions and determining the functions by minimizing the expansion's residual on the observation data while considering the prior-knowledge constraint that an ocean field varies smoothly in space and time. To test the effectiveness of the EF method, we applied it to reconstruct the 14-year cloud-free SST data in the Gulf Stream region spanning from 24.5°N to 44°N and from 82.5°W to 54.5°W. The original data consist of the 0.025° × 0.025° gridded daily composite daytime SST products of the Moderate-Resolution Imaging Spectroradiometer on the Aqua Sun-synchronous satellite, with an annual data-missing rate fluctuating around 78% in the region. In addition, we validated the reconstructed data against in situ buoy measurements. The reconstructed SST data's accuracies are −0.11 °C ± 0.91 °C (bias ± standard deviation of error) and −0.12 °C ± 0.67 °C in the areas without and with satellite observations, respectively, which are slightly lower and higher than the gappy satellite SST products' accuracy of −0.14 °C ± 0.77 °C.**

***Index Terms*— Data gap, empirical function (EF), matrix factorization, ocean remote sensing, sea surface temperature (SST) image.**

I. Introduction

PASSIVE satellite remote sensing is a crucial means of obtaining vital information on the expansive oceans, but cloud cover can result in gaps in the infrared and visual band data acquired by satellites, impeding the effective utilization of this data. For example, sea surface temperature (SST) is estimated from the satellite-sensed water leaving radiances in the infrared spectrum bands under cloud-free conditions [1]. However, historical observations suggest a global ocean cloud-cover rate of about 70% [2], [3]. Therefore, the absence of reliable geophysical parameter data from passive ocean remote sensing has been a persistent and significant challenge for satellite-based ocean observations in the infrared and visual spectra. Ocean research usually requires nongappy gridded data.

Optimal interpolation (OI) is a commonly used method for solving the data gap problem. However, the OI method requires additional background data to complete gappy data [4], and the error covariance of the background data is also necessary. Another popular solution is the data interpolating empirical orthogonal function (DINEOF) method [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. Unlike the OI method, the DINEOF method does not require additional background data.

Artificial intelligence (AI) provides another key to the problem and has recently attracted researchers' attention. AI can recover incomplete data similar to those in the ocean field. This is evident in areas such as computer vision, where incomplete images can be reconstructed, or recommender systems, where incomplete matrices of movie preference data can be completed. There are two popular AI approaches to the problem: the deep learning technique and the low-rank principle.

The deep learning technique achieved remarkable success in incomplete image recovery. Therefore, recent studies have tried to apply the deep learning technique to fill incomplete ocean data [15], [16], [17], [18]. The excellent performance demonstrates the great potential of the deep learning technique in this aspect. However, the design of the deep neural networks was empirical, and their complex structures have massive parameters, requiring many sample data to train them. In addition, the black box of the networks also involves the development of interpretation approaches [19], [20], [21].

The low-rank principle in the machine learning field is another way to recover incomplete matrices [22], [23], [24]. The principle is a prior knowledge that many real-world data in the form of matrices, such as images of natural scenes and data of movie preferences, are usually approximately low rank [24]. The principle means that a small number of latent factors underlying the data can account for most of the variation in the data. Thus, it is possible to recover the

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incomplete data by fitting them with the low-rank structure of matrices. There are two typical ways to apply the principle. The nuclear norm of a matrix, the sum of its singular values, is the convex approximation of the matrix rank. Thus, one way is to minimize the nuclear norm of the incomplete matrix under some constraints to determine its unobserved part while keeping its observed part as far as possible [23]. The other way is matrix factorization, where a product of two matrices with a preset rank approximates the incomplete matrix [22]. Matrix factorization determines the two matrices by minimizing the approximation deviation from the observed part of the incomplete matrix under some constraints. This approach has been applied in various fields, including recommender systems and image processing, to recover incomplete or corrupted data.

The idea of matrix factorization has already been embodied in the DINEOF method [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. The DINEOF method iteratively performs singular value decomposition (SVD) or the equivalent eigenvalue decomposition on an incomplete matrix. Each iteration retains and composes K-leading SVD components of the matrix to update the unobserved part of the matrix. The SVD components are so-called empirical orthogonal functions (EOFs) in the ocean field. The DINEOF method selects the optimal K from its implementation with different Ks. The result of the DINEOF-method implementation with a smaller K can initialize the one with the next larger K. The SVD in the DINEOF method means the orthogonality constraint imposed on the EOFs. However, the orthogonality constraint does not help improve the accuracy of completing missing data.

The idea of matrix factorization also inspired our study. We bridged the matrix factorization and the space-time-separation function expansion of the field and then proposed a novel method: the empirical function (EF) method. The EF method determines the expansion functions by minimizing the residual on the observed part of the field, and the expansion functions complete the unobserved part. We call the functions EFs, as they are not specified aforehand but derived from the data itself. (Moreover, the EFs are used for function expansion, and "EFs" is also the acronym for "expansion functions.")

In contrast to complex deep learning models that are hard to interpret, the EF method is characterized by a straightforward concept: EFs, each of which is a product of space- and time-varying parts, expand the field. In other words, we can consider the field the superposition of the space-time-separation fields. Thus, the EF method recovers incomplete data in an interpretable and much simpler way.

Different from the SVD (or eigenvalue decomposition) iteration in the DINEOF method, the framework of the EF method is the target optimization. The framework allows the EF method to conveniently incorporate the prior knowledge of an ocean field varying smoothly in space and time as the constraint on the EFs in the form of penalty terms. As this article will show, the smoothness constraint helps improve accuracy.

We tested the EF method in the synthetic data, which was used to test the DINEOF method [5]. We also applied the EF method to fill the 14-year satellite SST data with a mean data-missing rate of 78% and validated the completed SST data against a large amount of buoy data. The test and validation results show that the EF method reconstructed the incomplete data precisely.

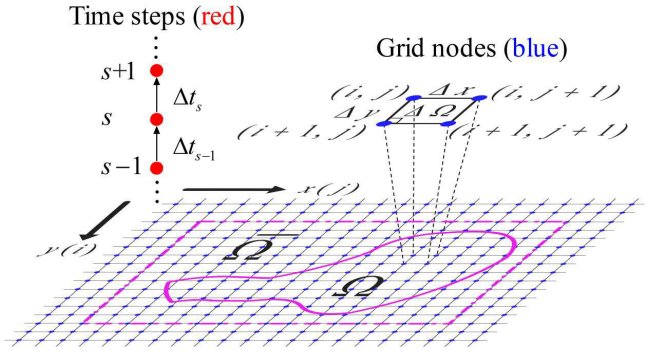


Fig. 1. Example to illustrate the extended regular rectangular spatial domain ($\bar{\Omega}$) of the irregular spatial domain (Ω) and the spatial–temporal discretization. The side lengths and area of a space grid are Δx, Δy, and $\Delta\Omega$. The interval between the sth and $(s+1)$th time steps is Δt_s.

Thus, we proposed a novel method, the EF method, which can precisely complete the gappy data of an ocean field in an interpretable and straightforward way. We organize the rest of this article as follows. Section II gives the EF method. Sections III and IV test the EF method using the synthetic data and the satellite and buoy SST data, respectively. Finally, Section V concludes this article.

II. Method

This section uses normal italic letters to represent scales, functions, or sets, and bold italic letters, matrixes, or vectors.

A. Description of the Incomplete Data Problem

Assume a spatial–temporal varying field (called the true field here and denoted by X^{t}), e.g., SST field, in a 2-D irregular spatial domain (denoted by Ω). The superscript "t" is the first letter of "true field."

Let x and y represent the two spatial dimensions. The irregular spatial domain Ω is discretized with rectangular grids (Fig. 1). For the convenience of the formulation, the irregular spatial domain Ω is extended to a regular rectangular domain conforming with rectangular grids. The extended spatial domain is denoted by $\bar{\Omega}$. i and j are the indexes of the grid node (i, j) in y- and x-dimensions.

Let t represent the temporal dimension discretized with time steps (Fig. 1). s is the index of the sth time step.

Assume that the field X^{t} is observed at some grid nodes and time steps during a period (denoted by T). The observations can be considered the values of an underlying field (called the observation field here and denoted by X^{o}) at the grid nodes and time steps. The superscript "o" is the first letter of "observation field."

Then, solving the incomplete data problem means constructing an approximate field in the spatial–temporal domain $\Omega \times T$ based on the observations. The symbol $\times$ denotes the Cartesian product between two sets. The approximate field is called the analysis field here and denoted by X^{a}. The superscript "a" is the first letter of "analysis field."

B. Index Mapping

The grid nodes in Ω can be reindexed in one dimension as $1, 2, \ldots, l, \ldots, L-1, L$, where L is the number of grid nodes in Ω. This mapping operation of a 2-D index to a 1-D index is denoted by f

$$l = f(i, j), \text{ where the node } (i, j) \text{ is in } \Omega. \tag{1}$$

Its inverse operation f^{-1} is

$$(i, j) = f^{-1}(l), \text{ where } l \in \{1, 2, \ldots, l, \ldots, L-1, L\}. \tag{2}$$

The above index mapping is introduced for converting 3-D data into a matrix.

With (1), the values $X^{\mathrm{t}}_{i,j,s}$, $X^{\mathrm{o}}_{i,j,s}$, and $X^{\mathrm{a}}_{i,j,s}$ of the fields X^{t}, X^{o}, and X^{a} at the node (i, j) and the time step s can be reindexed as $X^{\mathrm{t}}_{l,s}$, $X^{\mathrm{o}}_{l,s}$, and $X^{\mathrm{a}}_{l,s}$. Then, $X^{\mathrm{t}}_{l,s}$s, $X^{\mathrm{o}}_{l,s}$s, and $X^{\mathrm{a}}_{l,s}$s at all grid nodes and time steps in the spatial–temporal domain $\Omega \times T$ constitute three $L \times S$ matrixes (denoted by $\boldsymbol{X}^{\mathrm{t}}$, $\boldsymbol{X}^{\mathrm{o}}$, and $\boldsymbol{X}^{\mathrm{a}}$, respectively), where S is the number of time steps in T.

The matrixes $\boldsymbol{X}^{\mathrm{t}}$, $\boldsymbol{X}^{\mathrm{o}}$, and $\boldsymbol{X}^{\mathrm{a}}$ are discrete representations of the fields X^{t}, X^{o}, and X^{a}, while the matrix elements in $\boldsymbol{X}^{\mathrm{t}}$, $\boldsymbol{X}^{\mathrm{o}}$, and $\boldsymbol{X}^{\mathrm{a}}$ are the values of X^{t}, X^{o}, and X^{a} at the grid nodes and time steps in the spatial–temporal domain $\Omega \times T$. The row/column index of a matrix element in the matrixes specifies the matrix element's grid node/time step.

C. EF Expansion

If the rank of the matrix $\boldsymbol{X}^{\mathrm{t}}$ equals R, then we can exactly express $\boldsymbol{X}^{\mathrm{t}}$ as a product of an $L \times R$ matrix $\boldsymbol{U}^{\mathrm{t}}$ and an $S \times R$ matrix $\boldsymbol{V}^{\mathrm{t}}$

$$\boldsymbol{X}^{\mathrm{t}} = \boldsymbol{U}^{\mathrm{t}}\left(\boldsymbol{V}^{\mathrm{t}}\right)^{\mathrm{T}} \tag{3}$$

where the superscript "T" denotes matrix transpose.

If the rank of the matrix $\boldsymbol{X}^{\mathrm{t}}$ is greater than R, then (3) becomes an approximate equality.

Considering the above, we let

$$\boldsymbol{X}^{\mathrm{a}} = \boldsymbol{U}\boldsymbol{V}^{\mathrm{T}} \tag{4}$$

where $\boldsymbol{U}$ and $\boldsymbol{V}$ are $L \times R$ and $S \times R$ matrixes. $\boldsymbol{U}$ and $\boldsymbol{V}$ can be optimized by minimizing the difference between $\boldsymbol{X}^{\mathrm{a}}$ and $\boldsymbol{X}^{\mathrm{t}}$ in the least squares sense

$$\min_{\boldsymbol{U},\boldsymbol{V}} \|\boldsymbol{U}\boldsymbol{V}^{\mathrm{T}} - \boldsymbol{X}^{\mathrm{t}}\|_{\mathrm{F}}^2 \tag{5}$$

where $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm of a matrix.

However, we do not know $\boldsymbol{X}^{\mathrm{t}}$ but only a part of $\boldsymbol{X}^{\mathrm{o}}$. Therefore, instead of (5), we use

$$\min_{\boldsymbol{U},\boldsymbol{V}} \|\boldsymbol{P} \circ \left(\boldsymbol{U}\boldsymbol{V}^{\mathrm{T}} - \boldsymbol{X}^{\mathrm{o}}\right)\|_{\mathrm{F}}^2 \tag{6}$$

where $\circ$ denotes the Hadamard product between two matrixes. The matrix $\boldsymbol{P}$ is defined as

$$P_{l,s} = \begin{cases} 1, & \text{has observation at the} \\ & \text{node } f^{-1}(l) \text{and time step } s \\ 0, & \text{else.} \end{cases} \tag{7}$$

Let the grids and time steps be homogenous, and let their sizes approach zero. Then, formula (6) can be directly extended to the continuous case

$$\min_{\substack{U_1(x,y),\ldots,U_R(x,y),\\ V_1(t),\ldots,V_R(t)}} \iiint_{(\Omega\times T)\cap(P=1)} \left(\begin{matrix}\sum_r U_r(x,y)V_r(t)\\ -X^{\mathrm{o}}(x,y,t)\end{matrix}\right)^2 dt d\Omega \tag{8}$$

$P(x, y, t)$ is the characteristic function of the spatial–temporal domain where the field $X^{\mathrm{t}}(x, y, t)$ is observed

$$P(x, y, t) = \begin{cases} 1, & \text{has observation} \\ 0, & \text{else.} \end{cases} \tag{9}$$

The summing index r runs from 1 to R in (8).

The spatial functions $U_1(x, y), \ldots, U_r(x, y), \ldots, U_R(x, y)$ and temporal functions $V_1(t), \ldots, V_r(t), \ldots, V_R(t)$ are optimized through (8). $\sum_r U_r(x, y)V_r(t)$ is the approximate field $X^{\mathrm{a}}(x, y, t)$ of $X^{\mathrm{t}}(x, y, t)$ estimated from the known part of $X^{\mathrm{o}}(x, y, t)$, i.e., $X^{\mathrm{o}}(x, y, t)|_{P(x,y,t)=1}$

$$X^{\mathrm{a}}(x, y, t) = \sum_r U_r(x, y)V_r(t). \tag{10}$$

We name the functions $U_r(x, y)V_r(t)|_{r=1,\ldots,R}$ the EFs. Unlike the EOFs, the EFs are not required to be orthogonal.

As shown above, the matrix factorization is extended to (8) in the continuous case. The prior knowledge of a data matrix usually being low rank [24] suggests that the expansion (10) by space-time-separation functions could efficiently approximate the true field requiring a few expansion terms.

D. Spatially Temporally Smoothness Constraint

Ocean fields (e.g., SST field) are usually smooth in space and time, and this prior knowledge is taken here to constrain the EF expansion $X^{\mathrm{a}}(x, y, t)$ of $X^{\mathrm{t}}(x, y, t)$.

Each EF $U_r(x, y)V_r(t)$ is a product of a spatial-varying part $U_r(x, y)$ and a temporal-varying part $V_r(t)$. Thus, the following metrics are used to measure the spatial and temporal smooth degrees of $X^{\mathrm{a}}(x, y, t)$, respectively:

$$\sum_r \iint_\Omega |\nabla U_r(x, y)|^2 d\Omega \tag{11}$$

$$\sum_r \int_T \left(\frac{d}{dt}V_r(t)\right)^2 dt. \tag{12}$$

Formulas (11) and (12) are introduced into (8) as penalty terms

$$\begin{aligned}\min_{\substack{U_1(x,y),\ldots,U_R(x,y),\\ V_1(t),\ldots,V_R(t)}} & \iiint_{(\Omega\times T)\cap(P=1)} \left(\begin{matrix}\sum_r U_r(x,y)V_r(t)\\ -X^{\mathrm{o}}(x,y,t)\end{matrix}\right)^2 dt d\Omega \\ & + \alpha \sum_r \iint_\Omega |\nabla U_r(x, y)|^2 d\Omega \\ & + \beta \sum_r \int_T \left(\frac{d}{dt}V_r(t)\right)^2 dt \end{aligned} \tag{13}$$

where α and β are hyperparameters.

E. Calculation of EF Expansion With Spatially Temporally Smoothness Constraint

Formula (13) can be solved by using different representations of the field in principle, e.g., discretization representations with various spatial–temporal grids. However, we use rectangular grids again for their simplicity.

To unify the formulation in both cases of involving and not involving the boundary grid nodes and time steps of $\Omega \times T$, we define

$$\phi_{i,j} = \begin{cases} 1, & \text{Node } (i,j) \in \Omega \\ 0, & \text{Node } (i,j) \notin \Omega \end{cases} \tag{14}$$

$$\psi_s = \begin{cases} 1, & \text{Time step } s \in T \\ 0, & \text{Time step } s \notin T. \end{cases} \tag{15}$$

Taking (14) and (15) and adopting the second-order central difference, we get the approximations of three terms in (13)

$$\begin{aligned} &\iiint_{(\Omega\times T)\cap(P=1)} \left(\sum_r U_r(x,y)V_r(t) - X^{\mathrm{o}}(x,y,t)\right)^2 dtd\Omega \\ &\approx \sum_{l,s} P_{l,s}\left(\sum_r U_{l,r}V_{s,r} - X^{\mathrm{o}}_{l,s}\right)^2 \cdot \Delta\Omega \cdot \frac{\Delta t_{s-1}+\Delta t_s}{2} \end{aligned} \tag{16}$$

$$\begin{aligned} &\sum_r \iint_\Omega |\nabla U_r(x,y)|^2 d\Omega \\ &\approx \sum_r \begin{pmatrix} \frac{\Delta\Omega}{\Delta x^2}\sum_{i,j}\phi_{i,j+1}\phi_{i,j}\left(U_{i,j+1,r}-U_{i,j,r}\right)^2 \\ +\frac{\Delta\Omega}{\Delta y^2}\sum_{i,j}\phi_{i+1,j}\phi_{i,j}\left(U_{i+1,j,r}-U_{i,j,r}\right)^2 \end{pmatrix} \end{aligned} \tag{17}$$

$$\begin{aligned} &\sum_r \int_T \left(\frac{d}{dt}V_r(t)\right)^2 dt \\ &\approx \sum_r\sum_s \frac{1}{\Delta t_s}\psi_{s+1}\psi_s\left(V_{s+1,r}-V_{s,r}\right)^2 \end{aligned} \tag{18}$$

where Δx and Δy are the side lengths of a rectangular grid along the two spatial dimensions and $\Delta\Omega = \Delta x\Delta y$ is the area of the grid. Δt_s is the interval between the sth and $(s+1)$th time steps. $U_{i,j,r}$ and $U_{l,r}$ are the values of $U_r(x,y)$ at the grid nodes (i,j) and $f^{-1}(l)$, and $V_{s,r}$ is the value of $V_r(t)$ at the time step s.

After substituting (16)–(18), formula (13) becomes approximately

$$\min_{\boldsymbol{U},\boldsymbol{V}} \begin{pmatrix} \sum_{l,s} P_{l,s}\left(\sum_r U_{l,r}V_{s,r} - X^{\mathrm{o}}_{l,s}\right)^2 \cdot \Delta\Omega \cdot \frac{\Delta t_{s-1}+\Delta t_s}{2} \\ +\alpha\sum_r \begin{pmatrix} \frac{\Delta\Omega}{\Delta x^2}\sum_{i,j}\phi_{i,j+1}\phi_{i,j}\left(U_{i,j+1,r}-U_{i,j,r}\right)^2 \\ +\frac{\Delta\Omega}{\Delta y^2}\sum_{i,j}\phi_{i+1,j}\phi_{i,j}\left(U_{i+1,j,r}-U_{i,j,r}\right)^2 \end{pmatrix} \\ +\beta\sum_r\sum_s \frac{1}{\Delta t_s}\psi_{s+1}\psi_s\left(V_{s+1,r}-V_{s,r}\right)^2 \end{pmatrix}. \tag{19}$$

For short, we denote (19) as $\min_{\boldsymbol{U},\boldsymbol{V}}(Q(\boldsymbol{U},\boldsymbol{V})+\alpha J(\boldsymbol{U})+\beta G(\boldsymbol{V}))$.

The elements in the $\boldsymbol{U}$ and $\boldsymbol{V}$ parts of the gradient of the target function $Q(\boldsymbol{U},\boldsymbol{V})+\alpha J(\boldsymbol{U})+\beta G(\boldsymbol{V})$ are shown at the bottom of the next page.

Formula (19) can be iteratively optimized by gradient-based optimization methods, such as the classical formula of the kth iterative step in the gradient descent method

$$\begin{aligned} &\left(\boldsymbol{U}^{(k)},\boldsymbol{V}^{(k)}\right) \\ &= \left(\boldsymbol{U}^{(k-1)},\boldsymbol{V}^{(k-1)}\right) - \rho^{(k)} \\ &\quad\cdot(\nabla_{\boldsymbol{U}},\nabla_{\boldsymbol{V}})(Q(\boldsymbol{U},\boldsymbol{V})+\alpha J(\boldsymbol{U})+\beta G(\boldsymbol{V}))|_{\boldsymbol{U},\boldsymbol{V}=\boldsymbol{U}^{(k-1)},\boldsymbol{V}^{(k-1)}} \end{aligned} \tag{22}$$

where $\rho^{(k)}$ is the learning rate. The gradient required in the methods is calculated by (20) and (21), as shown at the bottom of the next page.

The estimation of $\boldsymbol{U}$ and $\boldsymbol{V}$ could be found by an existing data-completion method, e.g., the DINEOF method, to initialize the EF method. However, we give an optimization procedure independent of other data-completion methods: assume that the mean of the observed part of the matrix $\boldsymbol{X}^{\mathrm{o}}$ has been subtracted from the matrix elements in the observed part. First, the matrix elements in the unobserved part of $\boldsymbol{X}^{\mathrm{o}}$ are set to zero. $\boldsymbol{U}$ and $\boldsymbol{V}$ are initialized by the R-leading modes of the SVD of $\boldsymbol{X}^{\mathrm{o}}$ (or the equivalent eigenvalue decomposition). Second, $\boldsymbol{U}$ and $\boldsymbol{V}$ are optimized by the gradient descent method of a constant ρ several iterative times to find an initial estimation. In this article, the number of iterative times is set to 100. Finally, $\boldsymbol{U}$ and $\boldsymbol{V}$ are further optimized by the Adam method [25].

Taking the optimized $\boldsymbol{U}$ and $\boldsymbol{V}$, we get $\boldsymbol{X}^{\mathrm{a}} = \boldsymbol{U}\boldsymbol{V}^T$. By using the inverse index mapping (2), the elements of the matrix $\boldsymbol{X}^{\mathrm{a}}$ can be rearranged to the grid nodes and time steps to get the values of $X^{\mathrm{a}}(x,y,t)$ at the nodes and steps.

F. Division of Observation Data

Before optimizing (19), the observation data can be divided into two parts. Dividing data is equivalent to splitting the matrix $\boldsymbol{P}$ into two matrixes $\boldsymbol{P}_{\mathrm{opt}}$ and $\boldsymbol{P}_{\mathrm{val}}$

$$\boldsymbol{P} = \boldsymbol{P}_{\mathrm{opt}} + \boldsymbol{P}_{\mathrm{val}}. \tag{23}$$

The elements of the matrixes $\boldsymbol{P}_{\mathrm{opt}}$ and $\boldsymbol{P}_{\mathrm{val}}$ are 0 or 1. $P_{\mathrm{opt},l,s} = 1$ specifies the part of the observation data that will be used for optimizing (19). Meanwhile, $P_{\mathrm{val},l,s} = 1$ specifies the part of the observation data that will be used to evaluate the optimized $\boldsymbol{U}$ and $\boldsymbol{V}$ at each iterative step of the optimization procedure. The evaluation metric [root-mean-square error (RMSE)] for the kth iterative step is

$$\mathrm{RMSE}^{(k)} = \sqrt{\frac{\sum_{l,s} P_{\mathrm{val},l,s}\cdot\left(\sum_r U^{(k)}_{l,r}V^{(k)}_{s,r} - X^{\mathrm{o}}_{l,s}\right)^2}{\sum_{l,s} P_{\mathrm{val},l,s}}}. \tag{24}$$

The final optimized $\boldsymbol{U}$ and $\boldsymbol{V}$ are $\boldsymbol{U}^{(\arg\min_k \mathrm{RMSE}^{(k)})}$ and $\boldsymbol{V}^{(\arg\min_k \mathrm{RMSE}^{(k)})}$, having the minimum RMSE on the second part of the observation data.

For clarity, we call the first part the optimization observation data and the second part the validation observation data. In the tests of Sections III and IV, the ratio between the optimization and validation observation data is set to 9:1.

The optimization procedure is summarized in Algorithm 1.

Algorithm 1 Implementation of the EF Method

Main inputs: $\boldsymbol{X}^{\mathrm{o}}$, R, ρ, α, β, K_{GD}, K_{Adam}, N
Output: $\boldsymbol{X}^{\mathrm{a}}$

Set the unobserved matrix elements of $\boldsymbol{X}^{\mathrm{o}}$ to zero.
Generate the matrix $\boldsymbol{P}$ according to (7).
Split the matrix $\boldsymbol{P}$ into two matrices $\boldsymbol{P}_{\mathrm{opt}}$ and $\boldsymbol{P}_{\mathrm{val}}$ according to (23).
Generate $\boldsymbol{U}^{(0)}$ and $\boldsymbol{V}^{(0)}$ based on the R-leading SVD components of $\boldsymbol{X}^{\mathrm{o}} \circ \boldsymbol{P}_{\mathrm{opt}}$.
For $k = 1, \ldots, K_{\mathrm{GD}}$
  Calculate $\nabla_{\boldsymbol{U}} (Q + \alpha J + \beta G)|_{\boldsymbol{U},\boldsymbol{V}=\boldsymbol{U}^{(k-1)},\boldsymbol{V}^{(k-1)}}$ and $\nabla_{\boldsymbol{V}} (Q + \alpha J + \beta G)|_{\boldsymbol{U},\boldsymbol{V}=\boldsymbol{U}^{(k-1)},\boldsymbol{V}^{(k-1)}}$, using (20) and (21) where $\boldsymbol{P}$ is replaced with $\boldsymbol{P}_{\mathrm{opt}}$.
  Update $\boldsymbol{U}^{(k)}$ and $\boldsymbol{V}^{(k)}$ by the gradient descent method with the learning rate of ρ.
End
$\boldsymbol{U}^{(0)} \leftarrow \boldsymbol{U}^{(K_{\mathrm{GD}})}$, $\boldsymbol{V}^{(0)} \leftarrow \boldsymbol{V}^{(K_{\mathrm{GD}})}$, where "$\leftarrow$" denotes an assignment operation.
Calculate $\mathrm{RMSE}^{(0)}$ according to (24), and $\mathrm{RMSE}^{*} \leftarrow \mathrm{RMSE}^{(0)}$, $\boldsymbol{U}^{*} \leftarrow \boldsymbol{U}^{(0)}$, $\boldsymbol{V}^{*} \leftarrow \boldsymbol{V}^{(0)}$.
For $k = 1, \ldots, K_{\mathrm{Adam}}$
  Calculate $\nabla_{\boldsymbol{U}} (Q + \alpha J + \beta G)|_{\boldsymbol{U},\boldsymbol{V}=\boldsymbol{U}^{(k-1)},\boldsymbol{V}^{(k-1)}}$ and $\nabla_{\boldsymbol{V}} (Q + \alpha J + \beta G)|_{\boldsymbol{U},\boldsymbol{V}=\boldsymbol{U}^{(k-1)},\boldsymbol{V}^{(k-1)}}$, using (20) and (21) where $\boldsymbol{P}$ is replaced with $\boldsymbol{P}_{\mathrm{opt}}$.
  Update $\boldsymbol{U}^{(k)}$ and $\boldsymbol{V}^{(k)}$ by the Adam method with the learning rate of ρ.
  Calculate $\mathrm{RMSE}^{(k)}$ according to (24).
  If $\mathrm{RMSE}^{(k)} < \mathrm{RMSE}^{*}$
    $\mathrm{RMSE}^{*} \leftarrow \mathrm{RMSE}^{(k)}$, $\boldsymbol{U}^{*} \leftarrow \boldsymbol{U}^{(k)}$, $\boldsymbol{V}^{*} \leftarrow \boldsymbol{V}^{(k)}$.
  End
  If $\mathrm{RMSE}^{(k-N)}, \ldots, \mathrm{RMSE}^{(k)}$ monotonically increase
    Exit loop.
End
End
$\boldsymbol{X}^{\mathrm{a}} \leftarrow \boldsymbol{U}^{*}(\boldsymbol{V}^{*})^{\mathrm{T}}$.

G. Spatial–Temporal Domain Decomposition

We take the idea of domain decomposition to convert a big-size incomplete data problem into several small-size ones, thus lowering our method's requirement on a computing resource.

The spatial–temporal domain $\bar{\Omega} \times T$ is decomposed into nonoverlapping rectangular subdomains. $L_a \times L'_b \times T_c$ denotes the subdomain (a, b, c), while L_a, L'_b, and T_c, the ranges of the subdomain (a, b, c) in the two spatial dimensions and the temporal dimension.

The subdomains are extended outward by ΔL and $\Delta L'$ in the two spatial dimensions and by ΔT in the temporal dimension. $(L_a + 2\Delta L) \times (L'_b + 2\Delta L') \times (T_c + 2\Delta T)$ denotes the extended subdomain (a, b, c), while $(L_a + 2\Delta L)$, $(L'_b + 2\Delta L')$, and $(T_c + 2\Delta T)$, the ranges of the extended subdomain (a, b, c) in the two spatial and temporal dimensions. The extended subdomain (a, b, c) is partially overlapped with its adjacent extended subdomains. The overlapping part is $(L_a + 2\Delta L) \times (L'_b + 2\Delta L') \times (T_c + 2\Delta T) - (L_a - 2\Delta L) \times (L'_b - 2\Delta L') \times (T_c - 2\Delta T)$ where $(L_a - 2\Delta L) \times (L'_b - 2\Delta L') \times (T_c - 2\Delta T)$ denotes the subdomain (a, b, c) reduced inward by ΔL and $\Delta L'$ in the two spatial dimensions and by ΔT in the temporal dimension. This overlapping part is designed to achieve a gradual transition between the analysis fields of the extended subdomain (a, b, c) and its neighbors.

We construct the analysis field $X^{\mathrm{a}}_{\mathrm{sub},a,b,c}(x, y, t)$ of the extended subdomain (a, b, c), based on the observations inside the extended subdomain (a, b, c). Furthermore, we extend the definition domain of $X^{\mathrm{a}}_{\mathrm{sub},a,b,c}(x, y, t)$ from the extended subdomain (a, b, c) to $\bar{\Omega} \times T$ and fill zeros outside the extended subdomain (a, b, c)

$$X^{\mathrm{a}}_{a,b,c}(x, y, t) = \begin{cases} X^{\mathrm{a}}_{\mathrm{sub},a,b,c}(x, y, t), & (x, y, t) \in (L_a + 2\Delta L) \\ & \times \left(L'_b + 2\Delta L'\right) \times (T_c + 2\Delta T) \\ 0, & (x, y, t) \in \bar{\Omega} \times T - (L_a + 2\Delta L) \\ & \times \left(L'_b + 2\Delta L'\right) \times (T_c + 2\Delta T) \end{cases} \tag{25}$$

where $X^{\mathrm{a}}_{a,b,c}(x, y, t)$ can be calculated for each extended subdomain in a parallel manner.

Next, we superpose $X^{\mathrm{a}}_{a,b,c}(x, y, t)$s of all extended subdomains. In the superposition, we use 3-D window functions to achieve a gradual transition between $X^{\mathrm{a}}_{a,b,c}(x, y, t)$s of the

$$\begin{aligned} &\left[\nabla_{\boldsymbol{U}} \left(Q(\boldsymbol{U}, \boldsymbol{V}) + \alpha J(\boldsymbol{U}) + \beta G(\boldsymbol{V})\right)\right]_{l,r} \\ &= \sum_s P_{l,s} \left(\sum_r U_{l,r} V_{s,r} - X^{\mathrm{o}}_{l,s} \right) \Delta\boldsymbol{\Omega} (\Delta t_{s-1} + \Delta t_s) V_{s,r} \\ &\quad + \alpha \begin{pmatrix} \left(\frac{2\Delta\boldsymbol{\Omega}}{\Delta x^2} \phi_{i,j+1} + \frac{2\Delta\boldsymbol{\Omega}}{\Delta x^2} \phi_{i,j-1} + \frac{2\Delta\boldsymbol{\Omega}}{\Delta y^2} \phi_{i+1,j} + \frac{2\Delta\boldsymbol{\Omega}}{\Delta y^2} \phi_{i-1,j} \right) U_{i,j,r} \\ - \frac{2\Delta\boldsymbol{\Omega}}{\Delta x^2} \phi_{i,j+1} U_{i,j+1,r} - \frac{2\Delta\boldsymbol{\Omega}}{\Delta x^2} \phi_{i,j-1} U_{i,j-1,r} \\ - \frac{2\Delta\boldsymbol{\Omega}}{\Delta y^2} \phi_{i+1,j} U_{i+1,j,r} - \frac{2\Delta\boldsymbol{\Omega}}{\Delta y^2} \phi_{i-1,j} U_{i-1,j,r} \end{pmatrix} \end{aligned} \tag{20}$$

$$\begin{aligned} &\left[\nabla_{\boldsymbol{V}} \left(Q(\boldsymbol{U}, \boldsymbol{V}) + \alpha J(\boldsymbol{U}) + \beta G(\boldsymbol{V})\right)\right]_{s,r} \\ &= \sum_l P_{l,s} \left(\sum_r U_{l,r} V_{s,r} - X^{\mathrm{o}}_{l,s} \right) \Delta\boldsymbol{\Omega} (\Delta t_{s-1} + \Delta t_s) U_{l,r} \\ &\quad + \beta \begin{pmatrix} \left(\frac{2}{\Delta t_s} \psi_{s+1} + \frac{2}{\Delta t_{s-1}} \psi_{s-1} \right) V_{s,r} \\ - \frac{2}{\Delta t_s} \psi_{s+1} V_{s+1,r} - \frac{2}{\Delta t_{s-1}} \psi_{s-1} V_{s-1,r} \end{pmatrix} \end{aligned} \tag{21}$$

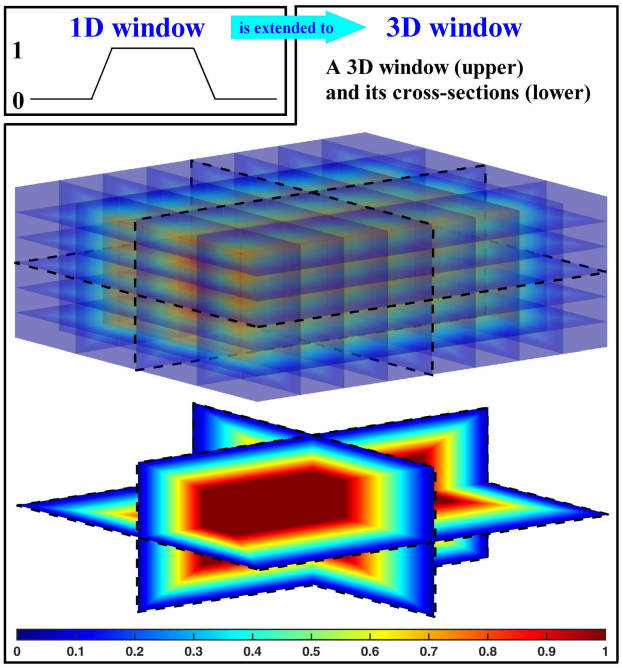


Fig. 2. Example of a 3-D trapezoidal window function. The 3-D trapezoidal window is the extension of the 1-D trapezoidal window in the 3-D case. The upper part of the 3-D-window panel shows a 3-D trapezoidal window, and its lower part shows the three cross sections of the 3-D trapezoidal window denoted by dashed lines in the upper part.

adjacent extended subdomains. The window function for the extended subdomain (a, b, c) can be expressed as

$$W_{a,b,c}(x,y,t) = \begin{cases} 1 & (x,y,t) \in \begin{array}{l}(L_a - 2\Delta L)\\ \times(L'_b - 2\Delta L')\\ \times(T_c - 2\Delta T)\end{array} \\ \text{smoothly changes} \\ \text{from 1 on} \\ \partial((L_a - 2\Delta L) & (x,y,t) \in \begin{array}{l}(L_a + 2\Delta L)\\ \times(L'_b + 2\Delta L')\\ \times(T_c + 2\Delta T)\\ -(L_a - 2\Delta L)\\ \times(L'_b - 2\Delta L')\\ \times(T_c - 2\Delta T)\end{array} \\ \times(L'_b - 2\Delta L') \\ \times(T_c - 2\Delta T)) \\ \text{to 0 on} \\ \partial((L_a + 2\Delta L) \\ \times(L'_b + 2\Delta L') \\ \times(T_c + 2\Delta T)) \\ \\ 0 & (x,y,t) \in \begin{array}{l}\bar{\Omega} \times T\\ -(L_a + 2\Delta L)\\ \times(L'_b + 2\Delta L')\\ \times(T_c + 2\Delta T)\end{array} \end{cases} \tag{26}$$

where $\partial(\cdot)$ denotes the boundary of the domain specified in the round bracket. Fig. 2 gives an example of a 3-D trapezoidal window function (used in the test of Section IV) for illustrative purposes, which is the extension of a 1-D trapezoidal window function in a 3-D case.

Finally, the analysis field of the spatial–temporal domain $(\bar{\Omega} \times T)$ is

$$X^{\mathrm{a}}(x,y,t) = \frac{\sum_{a,b,c} X^{\mathrm{a}}_{a,b,c}(x,y,t) \cdot W_{a,b,c}(x,y,t)}{\sum_{a,b,c} W_{a,b,c}(x,y,t)}. \tag{27}$$

We can get the analysis field of the concerned spatial–temporal domain $(\Omega \times T)$ included in $\bar{\Omega} \times T$.

The procedure of the spatial–temporal domain decomposition algorithm is summarized in Algorithm 2.

Algorithm 2 Implementation of the Spatial–Temporal Domain Decomposition

Main inputs: X^{o}, ΔL, $\Delta L'$, ΔT
Output: X^{a}

1: Decompose the spatial-temporal domain $\bar{\Omega} \times T$ into non-overlapping sub-domains.
2: Extend the sub-domains into the overlapping sub-domains outward by ΔL and $\Delta L'$ in the two spatial dimensions and by ΔT in the temporal dimension.
3: Use Algorithm 1 to generate the analysis field of each extended sub-domain, based on the part of X^{o} inside the extended sub-domain.
4: Generate $X^{\mathrm{a}}_{a,b,c}$ for each (a, b, c) where (a, b, c) is the sub-domain index, using (25).
5: Generate the window $W_{a,b,c}$ for each (a, b, c).
6: Generate the analysis field X^{a} of the domain $\bar{\Omega} \times T$, using (27).
7: Extract the analysis field X^{a} of the domain $\Omega \times T$ from that of the domain $\bar{\Omega} \times T$.

III. Synthetic Data Test

A spatial 1-D synthetic field is used to test the EF method, which was previously applied to test the DINEOF method [5]. The field is

$$\begin{aligned} X^{\mathrm{t}}(x,t) &= \sin\left(\frac{2\pi x}{L}\right)\sin\left(\frac{2\pi t}{S}\right) + \sin\left(\frac{4.2\pi x}{L}\right)\sin\left(\frac{4.2\pi t}{S}\right) \\ &+ \sin\left(\frac{6.2\pi x}{L}\right)\sin\left(\frac{6.2\pi t}{S}\right) + \tanh\left(\frac{2\pi x}{L}\right)\cos\left(\frac{2\pi t}{S}\right) \\ &+ \tanh\left(\frac{4\pi x}{L}\right)\cos\left(\frac{4.2\pi t}{S}\right) + \tanh\left(\frac{8\pi x}{L}\right)\cos\left(\frac{0.2\pi t}{S}\right) \\ &+ \tanh\left(\frac{4.8\pi x}{L}\right)\cos\left(\frac{2.2\pi t}{S}\right) + \tanh\left(\frac{2\pi x}{L} + \frac{2\pi t}{S}\right) \\ &+ \tanh\left(\frac{2\pi x}{L} + \frac{4\pi t}{S}\right). \end{aligned} \tag{28}$$

The discrete nodes of the 1-D space are $x = 0, 1, 2, \ldots, L-2, L-1$, while the time steps are $t = 0, 1, 2, \ldots, S-2, S-1$. In this section, both L and S are set to 101.

We tested the EF method in 12 settings of different signal-to-noise ratios (SNR) and data-missing rates in Table I. Take a setting of a 10-dB SNR and a 75% data-missing rate, for example. First, we generated the field data of (28) at the grid nodes and time steps as the ground truth shown in Fig. 3(a). Then, we added Gaussian noise to the field data. A 10-dB

TABLE I

DINEOF AND EF METHODS WERE USED TO COMPLETE THE OBSERVATION DATA IN 100 INSTANCES OF EACH SNR AND DATA-MISSING RATE SETTING, RESPECTIVELY. THE RMSE OF THE COMPLETION RESULT WAS CALCULATED IN EACH INSTANCE TO GET 100 RMSES FOR EACH METHOD UNDER EACH SETTING. THEN, THE STATISTICS OF 100 RMSES WERE FURTHER CALCULATED, INCLUDING 25% QUARTILE, MEDIAN (50% QUARTILE), 75% QUARTILE, AND TRI-MEAN

SNR	Data-missing rate	Statistics of RMSEs of DINEOF				Statistics of RMSEs of EF			
		25% quartile	Media	75% quartile	Tri-mean	25% quartile	Media	75% quartile	Tri-mean
30 dB	60%	0.0508	0.0555	0.0613	0.0558	0.0255	0.0261	0.0271	**0.0262**
	75%	0.0919	0.1003	0.1086	0.1003	0.0335	0.0352	0.0371	**0.0352**
	90%	0.6448	0.7402	0.8250	0.7375	0.0746	0.0799	0.0869	**0.0803**
20 dB	60%	0.1165	0.1209	0.1276	0.1214	0.0725	0.0737	0.0753	**0.0738**
	75%	0.1891	0.1977	0.2102	0.1987	0.0837	0.0851	0.0879	**0.0855**
	90%	0.7343	0.7995	0.8646	0.7995	0.1170	0.1234	0.1297	**0.1234**
10 dB	60%	0.3226	0.3295	0.3410	0.3306	0.1427	0.1489	0.1529	**0.1484**
	75%	0.4544	0.4686	0.4810	0.4682	0.1798	0.1870	0.1932	**0.1868**
	90%	0.9356	0.9656	1.0014	0.9670	0.3190	0.3333	0.3515	**0.3343**
5 dB	60%	0.5423	0.5504	0.5626	0.5514	0.2294	0.2368	0.2435	**0.2366**
	75%	0.7154	0.7390	0.7569	0.7376	0.2629	0.2722	0.2839	**0.2728**
	90%	1.0702	1.0890	1.1145	1.0907	0.3873	0.4065	0.4289	**0.4073**

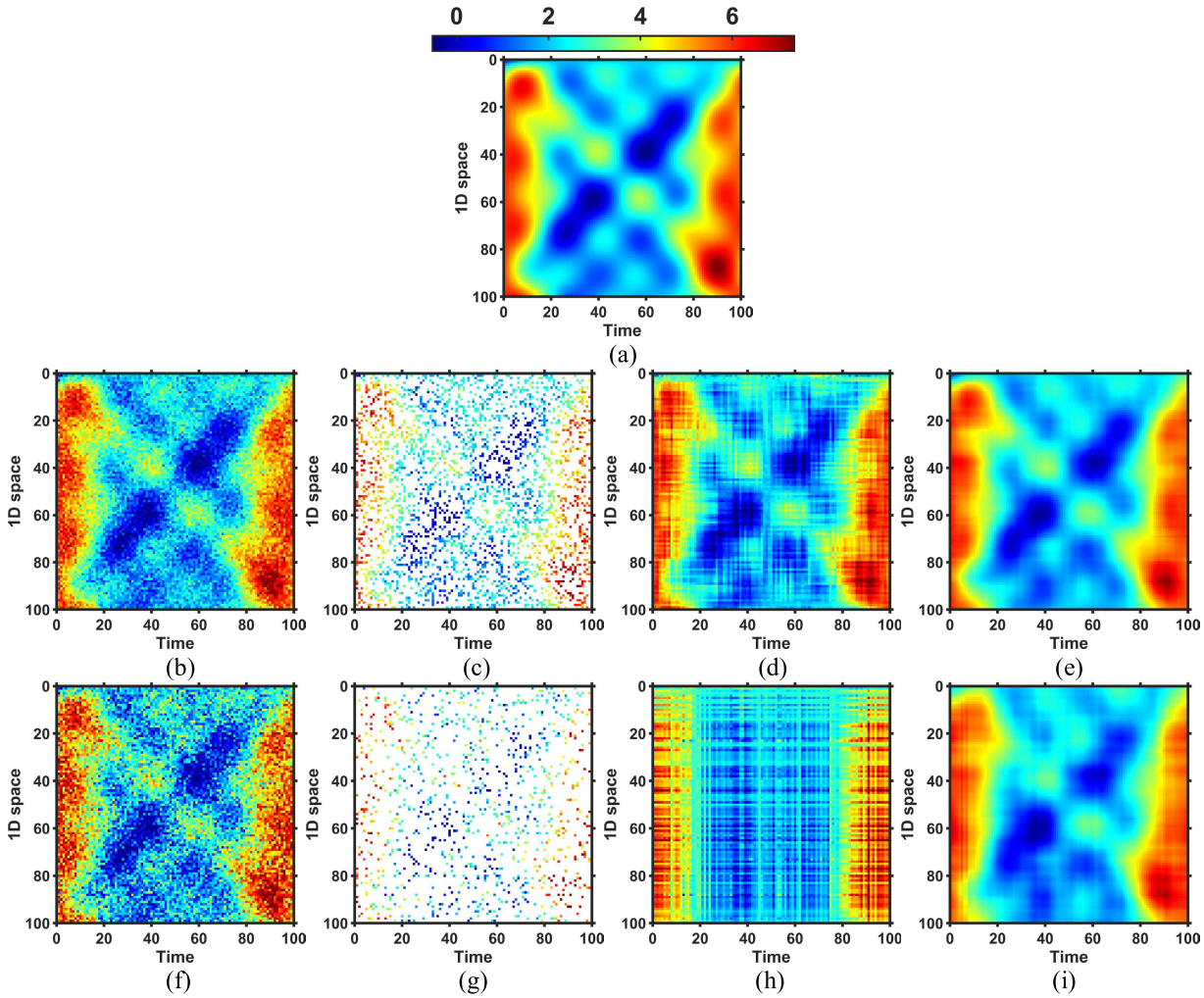


Fig. 3. Examples of synthetic data test. (a) Synthetic data. (b) Synthetic data added with 10-dB SNR Gaussian noise. (c) Partial observation of (b) under a 75% data-missing rate. (d) and (e) Completed data by the DINEOF method and the EF method, respectively, based on (c). (f)–(i) Same as (b)–(e) except for a 5-dB SNR and a 90% data-missing rate.

SNR means the ratio between the variance of the field data and the noise variance is 10 dB. Fig. 3(b) gives an instance of 10-dB SNR field data. A 75% data-missing rate means that the field (added noise) is observed at a random 25% of space nodes and time steps and is unobserved at the other 75%. Fig. 3(c) presents the observation of Fig. 3(b) under a 75%

data-missing rate with blanks where there is no observation. In the test, we set R and ρ to 100 and 0.005, respectively. We set both α and β to 5 for the 20- and 30-dB SNR settings and 50 for the 5- and 10-dB SNR settings, respectively.

Considering the randomness in the noise and unobserved sites, we generated 100 instances under every setting. We completed the observation data in each instance using the DINEOF and the EF methods (without domain decomposition). Fig. 3(d) and (h) shows the DINEOF method-completed data in an instance of a 10-dB SNR and a 75% data-missing rate [Fig. 3(c)] and that of a 5-dB SNR and a 90% data-missing rate [Fig. 3(g)], while Fig. 3(e) and (i) shows the EF method-completed data. Furthermore, we calculated the RMSE of the completed data in each instance and, thus, got 100 RMSEs for each method under each setting. Table I gives the 25%, 50% (median), and 75% quartiles, and tri-means of the 100 RMSEs for the DINEOF and EF methods under every setting. Fig. 3 and Table I show a significant improvement made by the EF method.

IV. Satellite Data Test

A. Test Area

The testing area spans from 24.5°N to 44°N and from 82.5°W to 54.5°W, wherein the noticeable feature observed in the SST field is the meandering Gulf Stream, recognized as one of the most robust western boundary ocean currents. The Gulf Stream transfers the warm water from the Florida Strait and flows up along the eastern coastline of the U.S. Then, it pulls away from the coast around Cape Hatteras, North Carolina, near 36°N, veers east, heads northeastward, and begins to meander more. The Gulf Stream makes this area have large SST spatial–temporal variation.

B. Satellite Data

The Aqua satellite, launched on May 4, 2002, has a Sun-synchronous orbit, crossing the equator at about 13:30 local time (LT) in its daytime passes. It carried the Moderate Resolution Imaging Spectroradiometer (MODIS) with a swath width of about 2330 km.

This study utilized data from the Aqua MODIS gridded daily daytime composite SST products provided by the National Oceanic and Atmospheric Administration (NOAA) CoastWatch (coastwatch.noaa.gov). To ensure higher quality cloud masks, CoastWatch specifically selected SSTs from daytime passes of the Aqua satellite and subsequently mapped them onto 0.025° × 0.025° grids using the arithmetic mean method. We collected the CoastWatch gridded satellite SST data in the test area for 14 years from January 1, 2006, to December 31, 2019, to test the EF method. Due to cloud cover, limited sensor coverage, and so on, many grid nodes and time steps have no valid SST data. Fig. 4(a) shows that the annual data-missing rate of the test area fluctuated around 78%. Second, the data-missing rate of 14 years was calculated within 0.5° × 0.5° grids to investigate the spatial distribution of missing data. Each 0.5° × 0.5° grid includes 400 0.025° × 0.025° grids of the gridded satellite SST data. Fig. 4(b) shows the data-missing rates over 0.5° × 0.5° grids range from 60% to 90%, where the grids with higher data-missing rates lie in the area near the Gulf Stream. Finally, we used the EF method to reconstruct the 0.025° × 0.025° gridded satellite SST data of the test area for 14 years. Section IV-D presents the validation results of the completed data against the NOAA buoy data.

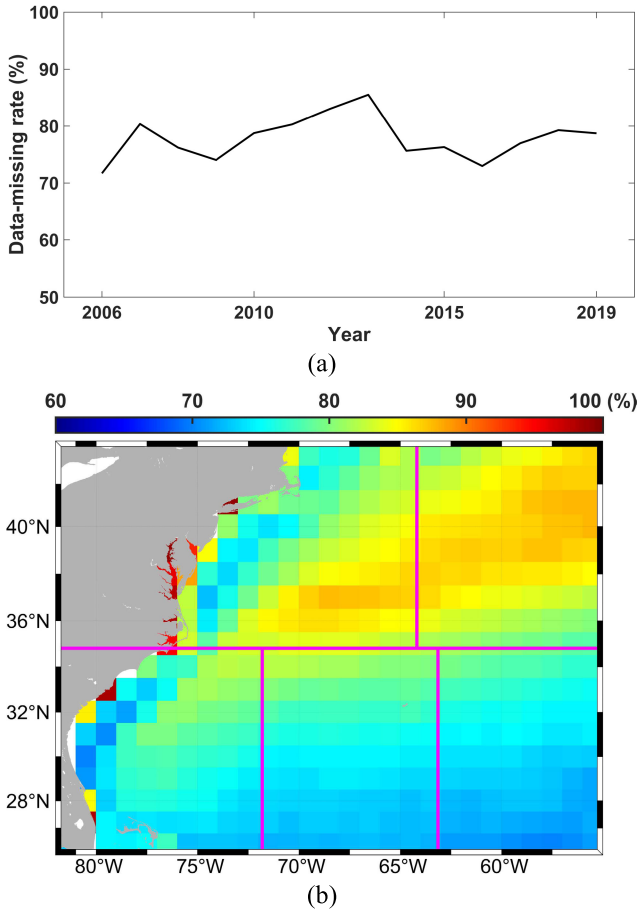


Fig. 4. Temporal variation and spatial distribution of the data-missing rate of the Aqua MODIS 0.025° × 0.025° gridded daily daytime composite SST products in the test area during 14 years (2006–2019). (a) Temporal variation. (b) Spatial distribution (that is data-missing rates of the SST products within 0.5° × 0.5° grids). The test area is divided into five subareas, and the magenta lines illustrate the boundaries of the subareas.

C. Reconstructing the Incomplete SST Data With the EF Method

According to Section II-G, we decomposed the spatial–temporal domain covering the test area in space and 14 years in time. First, we divided the test area into five nonoverlaid subareas [see Fig. 4(b)] and the 14 years into 14 nonoverlaid subperiods of one year. Then, we extended the subareas and the subperiods. Every two adjacent extended subareas have an overlaid strip area of a 60-grid width on their interface boundary. Meanwhile, every two extended subperiods have an overlaid period of 60 days at their interface end. Combining an extended subarea and an extended subperiod makes up an extended sub-spatial–temporal domain. Thus, we got 70 extended sub-spatial–temporal domains.

We completed the gridded satellite SST data extended sub-spatial–temporal domain by extended sub-spatial–temporal domain. We set R, ρ, α, and β to 300, 0.04,

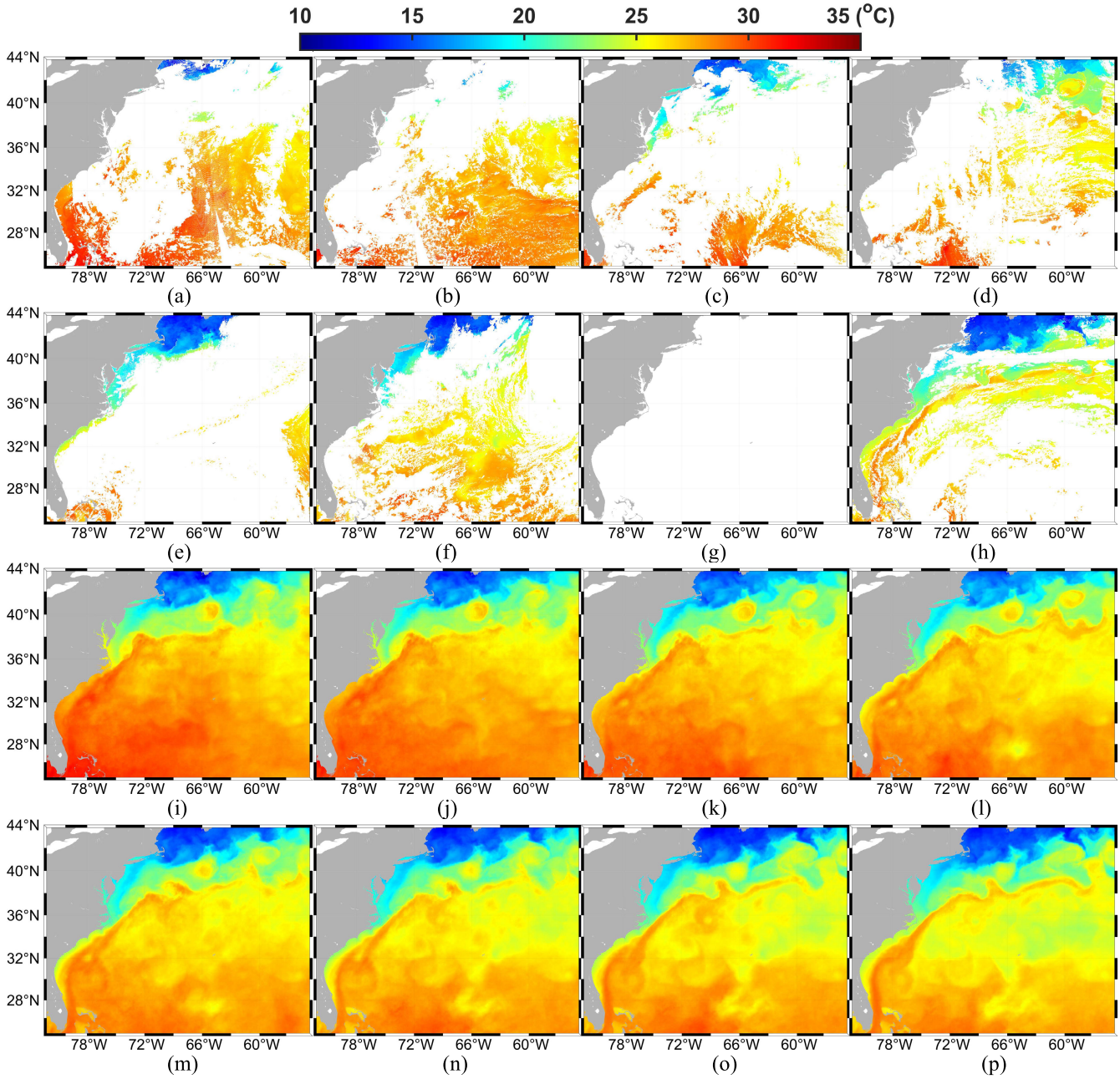


Fig. 5. Satellite and EF-completed SST data in three-day steps from 29 September to 20 October 2016. (a)–(h) Satellite SST data. (i)–(p) EF-completed SST data.

3.125E-2 degree^2·day, and 4.375 °C^2·degree^2·day^2 for all 70 extended sub-spatial–temporal domains, respectively.

Finally, we mosaiced the completed SST data of all extended sub-spatial–temporal domains together, according to Section II-G.

For the convenience of description, we classified the satellite SST data's grid nodes and time steps into the data-missing part (not having satellite data) and the non-data-missing part (having satellite data). After implementing the procedure in the above paragraphs, we completed the satellite SST data in the data-missing part while updating those in the non-data-missing part.

Fig. 5 presents the original gridded satellite SST data and the completed SST data in three-day steps from September 29 to October 20, 2016 (also see the video in the Supplemental Material showing the data in one-day steps). We can see that the SST field represented by the completed SST data varied smoothly without any noticeable abnormal abrupt changes. The completed SST data present well the morphological structure and evolution of the SST pattern of the Gulf Stream.

For comparison, we also completed the gridded satellite SST data of the extended sub-spatial–temporal domains by the DINEOF method and mosaiced them together. Fig. S1 in Supplemental Material presents the DINEOF-completed SST data on the days the same as Fig. 5. We can see that the DINEOF-completed SST data have some abrupt changes and more noise than the EF-completed SST data, and the DINEOF method failed to complete the SST data on the day without any satellite SST data [Fig. S1(f)].

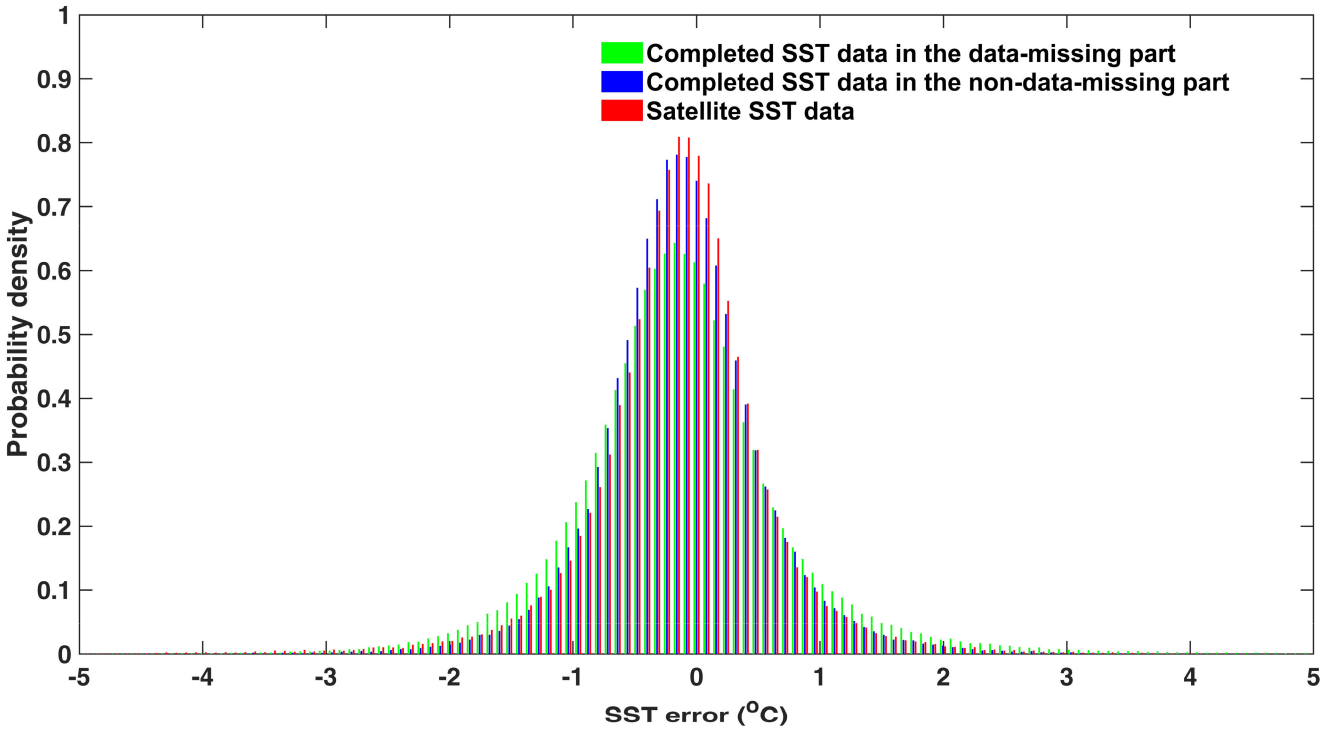


Fig. 6. Error probability distribution of the EF-completed SST data in the (satellite) data-missing and non-data-missing parts and the satellite SST data during 14 years (2006–2019).

D. Validation

To validate the accuracy of our data completion, we utilized the buoy data provided by the NOAA iQuam system (www.star.nesdis.noaa.gov/socd/sst/iquam/). The iQuam system is responsible for the quality control and monitoring of in situ SSTs, and it provides quality-controlled in situ SSTs with quality flags to various NOAA and external SST users. This is done in support of tasks such as satellite calibration and validation. The iQuam buoy data files contain in situ SSTs, observation locations and times, in situ platform types, and quality levels. Several studies have been conducted on the iQuam system in [26], [27], [28], [29], and [30].

After comprehensively considering the equatorial crossing time of the Aqua satellite in its daytime passes, the swath width of the Aqua MODIS, the latitude and longitude ranges of the test area, and the noise filtering of the in situ buoy SSTs, we only retained the in situ buoy SSTs within 13:35 LT $\pm$ 30 min and mapped them to the $0.025^\circ \times 0.025^\circ$ grids of the satellite SST data using the arithmetic mean method. Finally, we used the gridded buoy data as the ground truth and matched them with the completed SST data and the satellite SST data for validation.

We matched up 329 696 pairs of buoy SST data and completed SST data in the data-missing part, while 95 971 pairs of buoy SST data, completed SST data, and satellite SST data in the non-data-missing part. [The data-missing and non-data-missing parts account for 78% and 22% of the completed SST data, respectively. Thus, when we match the completed SST data with buoy SST data, the ratio between the sample size in the data-missing part and that in the non-data-missing part, which is 329 696:95 971 (3.4:1), is very close to 78%:22% (3.5:1).]

Using the gridded SST buoy data as the ground truth, we calculated the errors of the completed SST data in the data-missing and non-data-missing parts. We also calculated the errors of the satellite SST data in the non-data-missing part for comparison. Fig. 6 shows the error probability distributions of the completed SST data in the two parts, which have symmetrical bell shapes similar to the satellite SST data. In general, the errors of the completed SST data are slightly larger in the data-missing part than in the non-data-missing part. However, most errors of the completed SST data in both parts are minor within -1 °C–1 °C.

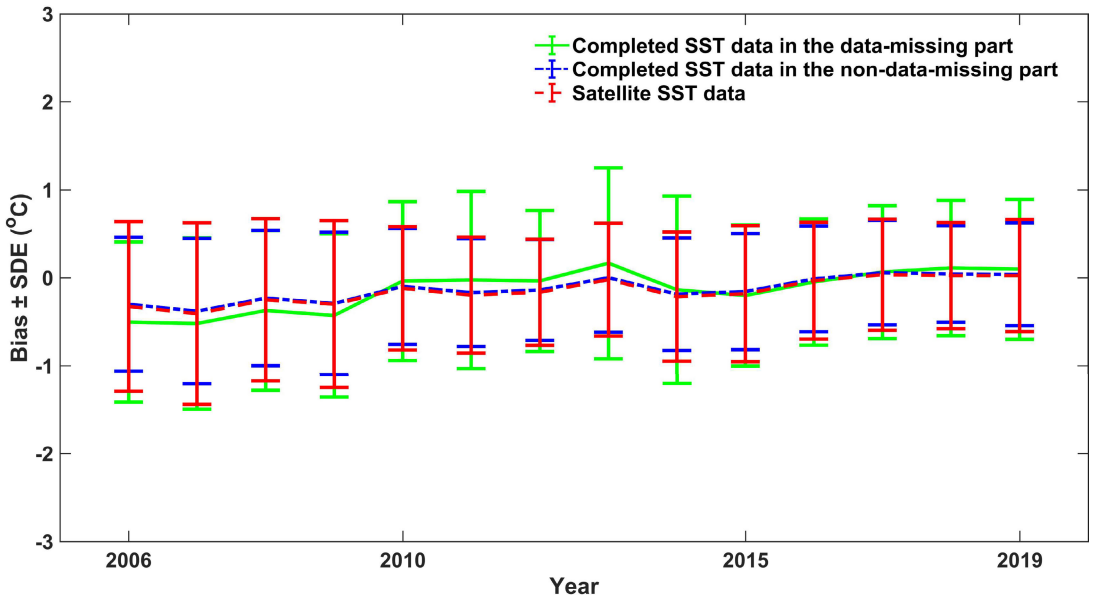
For quantitative comparison, we calculated the biases and standard deviations of error (SDEs) of the completed and satellite SST data for different SST ranges.

The biases for the whole SST range are -0.11 °C, -0.12 °C, and -0.14 °C for the completed SST data in the data-missing and non-data-missing parts and the satellite SST data, while the SDEs are 0.91 °C, 0.67 °C, and 0.77 °C. For comparison, we also calculated the bias and SDE of the DINEOF-completed SST data for the whole SST range, which are 0.35 °C and 2.81 °C in the data-missing part and -0.12 °C and 0.75 °C in the non-data-missing part. More details on the error of the DINEOF-completed SST data are given in Table S1 and Figs. S2–S4 of Supplemental Material, including the error distribution with respect to SST, error temporal variation, and error spatial distribution. The error distribution and variation comparison show that the EF method has higher accuracy than the DINEOF method. (The error statistics in different SST ranges, error temporal variation, and error spatial distribution of the EF method will be given later in Table II and Figs. 7 and 8.)

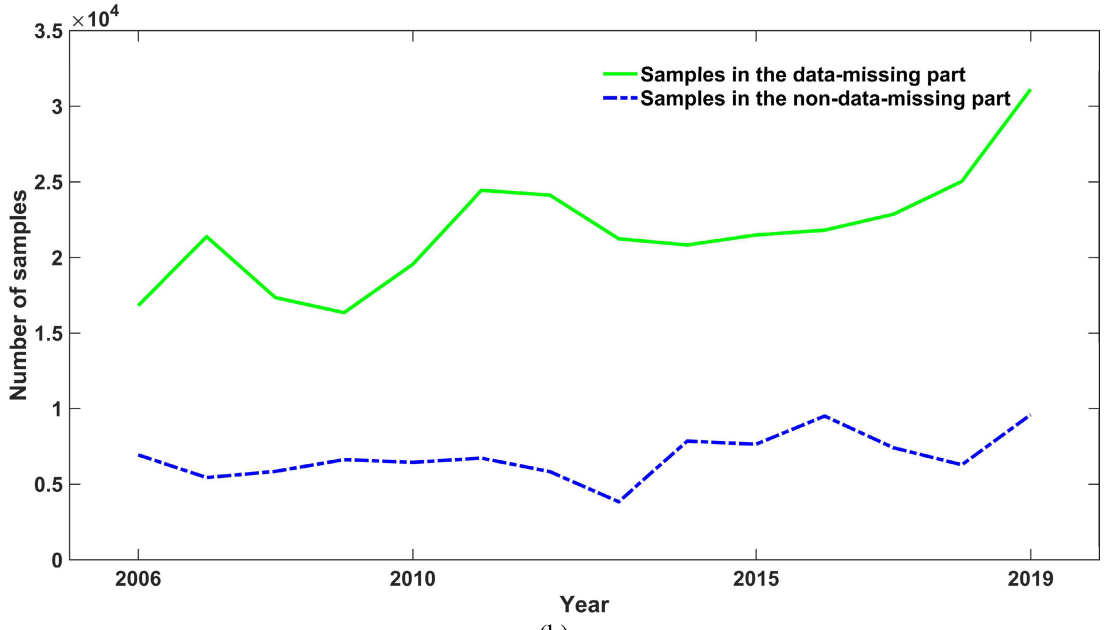
TABLE II

BIASES AND SDES OF THE WHOLE TEST AREA DURING 14 YEARS (2006–2019) IN DIFFERENT SST RANGES FOR THE EF-COMPLETED SST DATA IN THE (SATELLITE) DATA-MISSING AND NON-DATA-MISSING PARTS AND THE SATELLITE SST DATA

SST range (°C)	Error (Bias ± SDE, °C)		
	EF-completed data in the data-missing part	EF-completed data in the non-data-missing part	Satellite SST data
All	−0.11 ± 0.91	−0.12 ± 0.67	−0.14 ± 0.77
< 10	0.05 ± 1.10	0.01 ± 0.74	0.00 ± 0.85
10 to 20	0.20 ± 1.15	0.16 ± 0.79	0.12 ± 0.90
>20	−0.26 ± 0.70	−0.23 ± 0.59	−0.24 ± 0.69



(a)



(b)

Fig. 7. Bias and SDE temporal variations of the EF-completed SST data in the (satellite) data-missing and non-data-missing parts and the satellite SST data during 14 years (2006–2019). (a) Bias (curves) and SDE (bars) in each year. The biases and SDEs in a year were calculated from all samples in that year. (b) Number of samples in each year.

Table II shows the biases of the completed SST data have slight variation, positive in the lower and middle SST range (<10 °C and 10 °C–20 °C) but negative in the higher SST range (>20 °C). A similar bias variation also exists in the satellite SST data (see Table II). Thus, the bias variation should originate from the satellite SST data. Table II shows that the

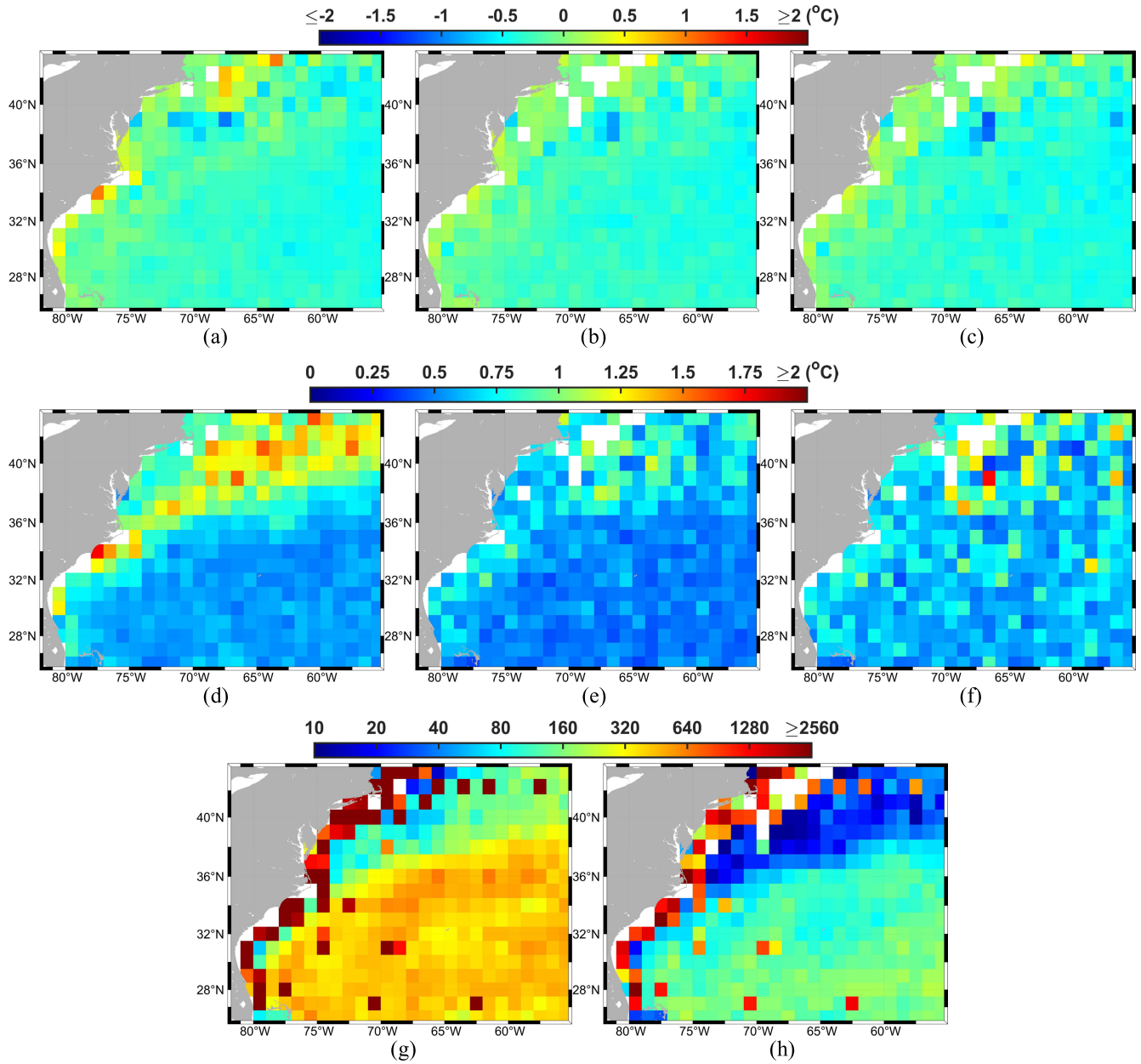


Fig. 8. Bias and SDE spatial distributions of the EF-completed SST data in the (satellite) data-missing and non-data-missing parts and the satellite SST data during 14 years (2006–2019). (a)–(c) Bias spatial distributions. (d)–(f) SDE spatial distributions. The biases and SDEs of the 0.025° × 0.025° EF-completed and satellite SST data during the 14 years were calculated within 0.5° × 0.5° grids. (g) Numbers of the samples during the 14 years within 0.5° × 0.5° grids in the data-missing part, while (f) those in the non-data-missing part.

SDEs of the completed SST data in the data-missing and non-data-missing parts are close to and lower than that of the satellite SST data for all different SST ranges, respectively. The accuracy improvement in the non-data-missing part could be due to the spatially temporally smoothness constraint in the EF method that smooths the noise in the satellite SST data.

To show the temporal variations of errors in the completed and satellite SST data, we calculated the annual bias and SDEs for the 14 years, respectively. The lines in Fig. 7(a) show the annual bias variations, while the bars on the lines are the annual SDEs. Fig. 7(b) gives the corresponding numbers of samples in the annual statistics. All annual bias and SDEs perform steadily in time with minor fluctuations. The annual SDEs are slightly greater in the completed SST data in the data-missing part than in the other data. In addition, all 14 annual SDEs decreased in the completed SST data in the non-data-missing part compared with those in the satellite SST data.

For investigating the spatial distributions of errors in the completed and satellite SST data, we calculate their biases and SDEs within 0.5° × 0.5° grids during the 14 years, respectively. Each 0.5° × 0.5° grid embraces 400 0.025° × 0.025° grids. We only calculate the biases and SDEs within the 0.5° × 0.5° grids with ≥10 samples. Fig. 8(a)–(c) shows the bias distributions in the completed SST data in the data-missing and non-data-missing parts and the satellite data, while Fig. 8(d)–(f) shows the SDE distributions. Fig. 8(g) and (h) shows the distributions of the numbers of samples in the data-missing and non-data-missing parts, respectively. The biases in all SST data are close to 0 °C in

TABLE III

BIAS AND SEDS OF THE WHOLE TEST AREA DURING 14 YEARS (2006–2019) FOR THE EF-COMPLETED SST DATA WITH NO CONSTRAINT, SPATIAL SMOOTHNESS CONSTRAINT, TEMPORAL SMOOTHNESS CONSTRAINT, AND TEMPORAL–SPATIAL SMOOTHNESS CONSTRAINT

	Error (Bias ± SDE, °C)	
	EF-completed data in the data-missing part	**EF-completed data in the non-data-missing part**
No constraint	1.21±5.74	−0.13±0.99
Spatial smoothness constraint	0.72±3.56	−0.14±0.75
Temporal smoothness constraint	−0.01±1.08	−0.12±0.71

most 0.5° × 0.5° grids, although the bias magnitudes in the completed SST data in the data-missing part are slightly larger in some places. Larger SDEs mainly lie near the Gulf Stream. Those larger SDEs could be caused by the spatial–temporal variation of the SST field being much sharper than in the other part of the test area. The SDEs are overall smaller in the completed SST data than in the satellite SST data in the non-data-missing part.

E. Ablation Experiment

We carried out an ablation experiment to investigate the effect of the spatial–temporal smoothness constraint. We completed the satellite data by the EF method without any constraint, with the spatial smoothness constraint, and with the temporal smoothness constraint. Thus, we set both α and β to zero, set α to zero but keep β unchanged, and set β to zero but keep α unchanged. Table III gives the error statistics in three cases. The EF method without any constraint has the largest error, and introducing the spatial or temporal smoothness constraints can reduce the error to varying degrees. In comparing the error statistics in Tables II and III, the EF method with the spatial–temporal smoothness constraint achieves the highest accuracy. Therefore, the spatial–temporal smoothness constraint benefits the data completion.

V. CONCLUSION

Due to cloud cover, ocean data acquired by satellites in the infrared and visual bands usually have gaps.

The major innovative contribution of our work is that we proposed a novel method, the EF method, for completing the gappy data of an ocean field, such as an SST field. The method can conveniently include prior knowledge, e.g., spatial–temporal smoothness constraint. The work was inspired by the idea of matrix factorization in the AI field and is summarized as follows.

First, by reindexing the grid nodes of the incomplete data of an ocean field in an irregular area, we convert the data into an incomplete matrix factorized as a product of two unknown matrixes. Then, we extended the matrix factorization to the formulas in the continuous case. The meaning of the formulas (the EF method) is intuitive that space-time-separation functions (EFs) expand the field. Each function is a product of unknown space- and time-varying parts. Our method determines two parts by minimizing the target function, the residual square sum of the expansion in the observation data. Considering the prior knowledge that an ocean field usually varies smoothly in space and time, the method requires two parts to be smooth. The minimization incorporates this smoothness constraint as penalty terms.

The EF method was tested in a large region, including the Gulf Stream, using gappy satellite SST data spanning 14 years (2006–2019). The completed SST data were validated against multiple buoy SST data. In the places without and with satellite observations, the accuracies of the completed SST data are $-0.11\ ^\circ\text{C} \pm 0.91\ ^\circ\text{C}$ (bias ± SDE) and $-0.12\ ^\circ\text{C} \pm 0.67\ ^\circ\text{C}$, which are slightly lower and higher than $-0.14\ ^\circ\text{C} \pm 0.77\ ^\circ\text{C}$ of the satellite SST products, respectively.

Moreover, the EF method could be extensible on the following aspects.

1) The EF method can be extended to other spatial–temporal dimensions.

2) While the EF method is implemented using rectangular grids in this article, the formulas presented in the continuous case can also be applied to other field representations, such as discretized representations with varying spatial grids and time steps. This flexibility enables deriving a range of algorithms for implementing the EF method.

3) Compared to the SVD (or eigenvalue decomposition) iteration used in the DINEOF method, the target function optimization framework of the EF method is more adaptable. It allows for including prior knowledge through penalty terms in the form of specific metrics within the target function.

Regarding its applications, the EF method proposed in this article is independent of data type. As a result, it can reconstruct various types of ocean data, not just SST data obtained from satellite images. This is particularly useful for addressing missing data caused by fog and rain, limited observation coverage due to satellite orbit and sensor swath width constraints, and data rejected by quality control measures.

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